Lecture 09 (B08): Constraints, Residual Kernels, Loss Design

Hard vs soft constraints, six loss kernels, data parallelism for DEQNs

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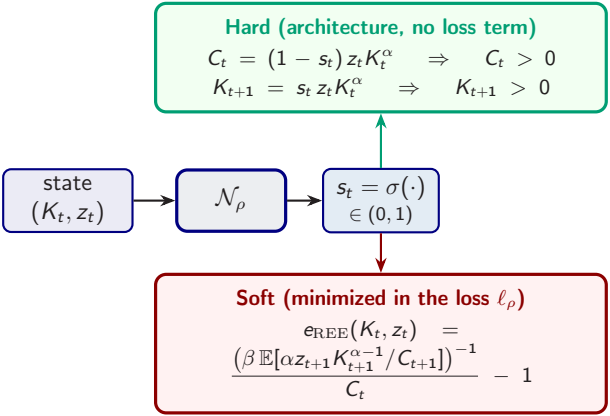
Brock–Mirman: Hard vs. Soft Constraints

Split the equilibrium conditions by what the architecture can enforce exactly.



Why bother? A softplus head on C_t alone would secure $C_t > 0$ but could push $K_{t+1} < 0$. The sigmoid-savings head removes *both* failure modes before training starts. Only the genuinely non-closed-form condition, the Euler equation, enters the loss.

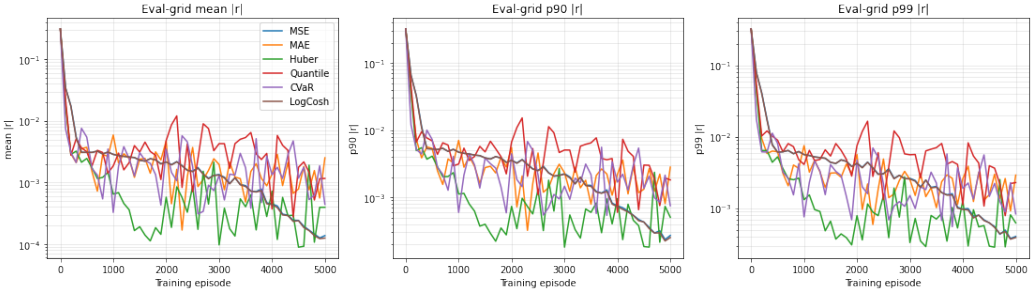
Choice of Training Loss: Six Kernels

The loss kernel is a modeling decision, not a default. Same Euler residual r, same network, same data, only the reduction $r \mapsto \ell$ changes:

Kernel	Formula	Character
MSE	$\frac{1}{B}\sum r^2$	quadratic; canonical default
MAE	$\frac{1}{B}\sum \lvert r\rvert$	constant gradient $\Rightarrow$ stalls at small $\lvert r\rvert$
Huber	quad. for $\lvert r\rvert \le \delta$, linear above	robust hybrid, but kink at $\lvert r\rvert = \delta$
Quantile	pinball at $\tau = 0.9$	trains the upper τ-quantile only
CVaR	mean of top 10% of $\lvert r\rvert$	explicitly trains the worst-case states
LogCosh	$\frac{1}{B}\sum \log\cosh(r)$	C^∞: $\frac{1}{2}r^2$ near 0, $\lvert r\rvert - \log 2$ in tails, no kink

Same Brock–Mirman model, full depreciation so $s^\star = \alpha\beta$ is in closed form; same network (2×32 swish, sigmoid head); same Adam optimizer; same CRN training stream. Six runs differ *only* in the reduction applied to the residual vector.

Loss Kernels: Convergence on Brock–Mirman



Mean / p_{90} / p_{99} of the absolute relative Euler error $|r|$ on a fixed evaluation grid, vs. training episode (log scale, 5000 episodes per kernel). LogCosh and MSE reach the lowest mean residual. Huber and CVaR produce the narrowest tail. MAE stalls at a noise floor.

Loss-Kernel Comparison: What to Take Away

- LogCosh converges most smoothly: the only C^∞ kernel that is MSE-like near $r = 0$ *and* MAE-like in the tails, no kink anywhere. Lowest-risk default when the residual distribution is unknown.
- MAE stalls at a noise floor: constant gradient magnitude prevents smaller steps as residuals shrink.
- Huber is robust but its kink at $|r|=\delta$ can introduce small oscillations.
- Quantile and CVaR narrow p_{99} at the cost of mean residual. Use them when worst-case policy quality matters (binding constraints, rare-but-consequential states, regulatory back-tests).
- Evaluate on the economic metric — Euler error along simulated paths, in CE-welfare-loss units — not on the training loss itself. The CE ranking is generally not the training-loss ranking.

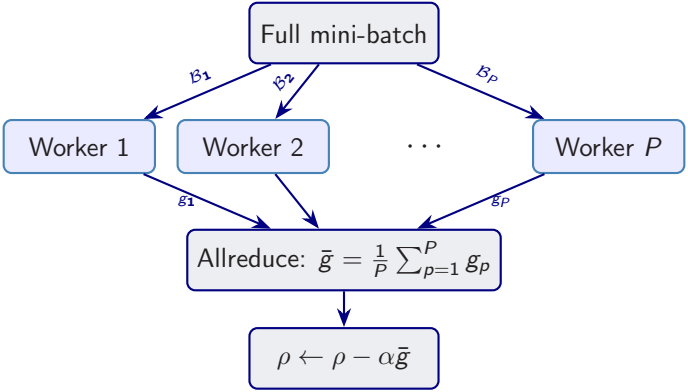
Practical default

MSE for smooth problems; **LogCosh** as a drop-in that is more forgiving with heavy-tailed residuals; **Huber** / **Quantile** / **CVaR** when the worst-case state matters.

Companion notebook: `lecture_09_B08_05_StochasticBM_LossComparison.ipynb`.

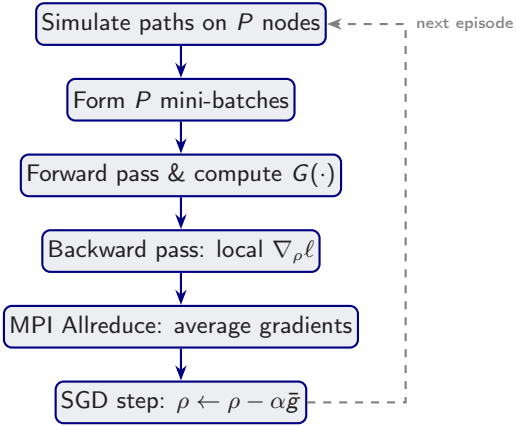
Data Parallelism for DEQNs

Idea: distribute training data across multiple workers (GPUs/CPUs).



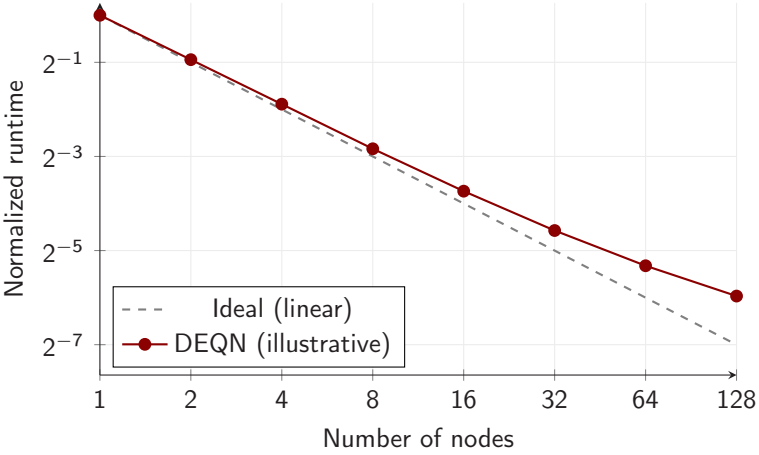
Implemented via **Horovod**/MPI. Each worker computes a local gradient; allreduce averages them.

DEQN Parallelization Scheme



Communication cost $\sim \mathcal{O}(|\rho|)$ per step, independent of data size.

Illustrative Scaling Pattern



Schematic message only: synchronous data parallelism is often close to linear for moderate worker counts; large clusters become increasingly communication-limited.

Hands-on Notebooks

Four notebooks accompany this lecture:

1. `01_Brock_Mirman_1972_DEQN.ipynb`
   Deterministic Brock–Mirman, exogenous uniform sampling of states.
2. `02_Brock_Mirman_Uncertainty_DEQN.ipynb`
   Stochastic Brock–Mirman ($\varrho > 0$, $\sigma_z > 0$), simulation-based sampling on the ergodic set.
3. `03_DEQN_Exercises_Blanks.ipynb` (morning exercises, blank template for student populate)
4. `04_DEQN_Exercises_Solutions.ipynb` (morning exercises, solved)

Notebooks 01–02 illustrate the sampling progression: uniform random $\rightarrow$ simulated trajectories. Notebooks 03–04 add KKT multipliers + Fischer–Burmeister + the 6-period OLG structure previewed on the previous slide.

All notebooks are on Nuvolos Cloud and the course GitHub repository.

Questions?

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https://sischei.github.io

Materials: https://github.com/sischei/Deep_Learning_for_Solving_And_Estimating_Dynamic_Economic_Models

Next up:

DEQN Exercises

10:45–12:00